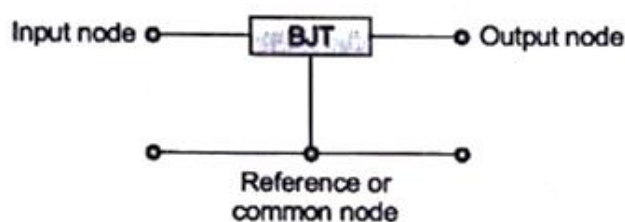


BJT Configuration:



Based on the reference node a BJT can be used in 3 configuration as given in table below:

Configuration	Input node	Output node
Common Base (CB)	E	C
Common Emitter (CE)	B	C
Common Collector (CC)	B	E

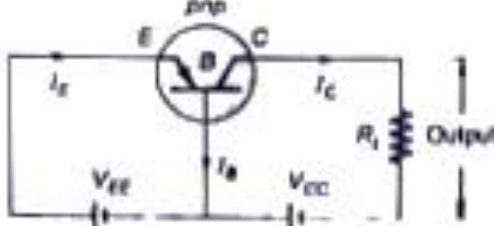
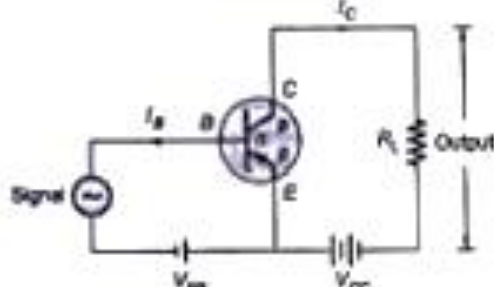
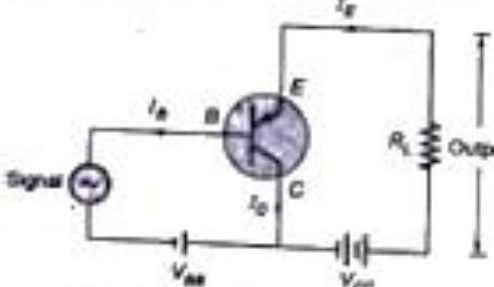
Transistor Sections:

- Emitter:** Heavily doped as comparison to other regions. The main function of this region is to supply majority charge carrier i.e. either electrons or holes.
- Base:** This section is very lightly doped and very thin as compared to other section. Its main function is to transfer the majority carriers from emitter to collector.
- Collector:** This section is moderately doped and has largest area than other two regions because it collects the charge carries from emitter and base. There is a large amount of heat liberation and so it is provided with large area to dissipate the heat.

Voltage relations	NPN Mode	PNP Mode
$V_E < V_B < V_C$	Active	Reverse
$V_E < V_B > V_C$	Saturation	Cutoff
$V_E > V_B < V_C$	Cutoff	Saturation
$V_E > V_B > V_C$	Reverse	Active

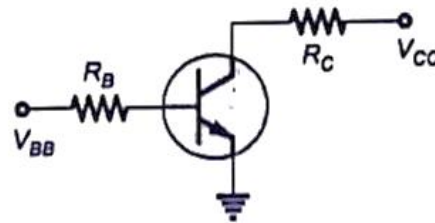
Mode of Operation:

Mode	Emitter-Base (E-B) Junction	Collector-Base (C-B) Junction	Properties	Application
Cut-off	Reverse bias	Reverse bias	Very high internal resistance	OFF-switch
Active	Forward bias	Reverse bias	Excellent transistor action	Amplifier
Saturation	Forward bias	Forward bias	Very low internal resistance	ON-switch
Reverse Active	Reverse bias	Forward bias	Very poor transistor action	Attenuator (Practically not used)

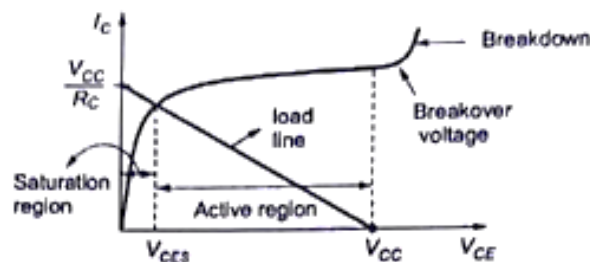
Configuration	Current Amplification factors
 Common Base p-n-p transistor	Current amplification factor (α) When no signal is applied then the ratio of the collector current to emitter current is called current amplification factor or current gain, α of a transistor is a measure of the quality of transistor. Higher is the value of α better is the transistor as I_C approaches I_E.
 Common emitter p-n-p transistor	Base current amplification factor (β) When no signal is applied the the ratio of collector current to the base current is called β. β usually ranges from 20 to 500. Common emitter configuration is used when appreciable current gain as well as voltage gain is required.
 Common collector p-n-p transistor	Current amplification factor (γ) When no signal is applied then the ratio of emitter current to the base current is called γ of the transistor. Common collector configuration has very high input resistance $\approx 750 \text{ k}\Omega$ and very low output resistance. Voltage gain is always less than one, so never used for amplification. This configuration is used for impedance matching i.e. driving a low impedance load from a high impedance source.

DC Load Lines

DC load line is the locus of all possible operating point at which BJT remains in active region.



- If base current $I_B < \frac{I_{Csat}}{\beta}$, then the transistor is being operated in active region.



- If base current $I_B > \frac{I_{Csat}}{\beta}$, then the transistor is being operated in saturated region.
- Collector current, $I_C = \frac{V_{CC} - V_{CE}}{R_C}$ for above figure.
- At saturation, $V_{CE} = V_{CES}$
- $V_{CE} = V_{CC}$ at cut-off when $I_C = 0$.
- Maximum value of current, $I_{Cmax} = \frac{V_{CC}}{R_C}$ in ideal case. [Taking $V_{CE} = 0$ at saturation]
- Power dissipated in a transistor $P_D = I_C V_{CE}$
- Power dissipation is maximum in active region and minimum in cut-off and saturation region.

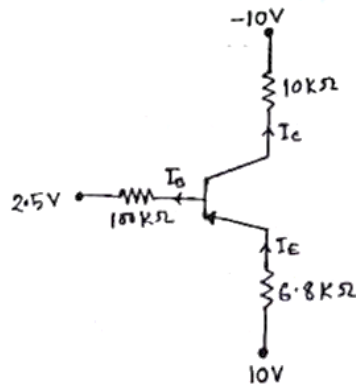
BJT Current-Voltage Relationship in Active Mode		
$\beta = \frac{I_C}{I_B}$	$\Rightarrow I_C = \beta I_B$	I_B = Base Current
$\alpha = \frac{I_C}{I_E}$		I_C = Collector Current
$I_E = I_C + I_B$		I_E = Emitter Current
$= (\beta + 1) I_B$		I_S = Saturation Current
		V_{BE} = Base-Emitter Voltage
$I_C = I_S e^{\frac{V_{BE}}{V_T}}$	for n-p-n transistor	V_T = Thermal voltage = $\frac{kT}{q}$
$I_C = I_S e^{\frac{V_{EB}}{V_T}}$	for p-n-p transistor	≈ 25 mV at room temperature

BJT Bias Configurations

Type	Configuration	Pertinent Equations	
Fixed-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C R_C$	
Emitter-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $R_i = (\beta + 1)R_E$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$	
Voltage-divider bias		EXACT: $R_{Th} = R_1 R_2, E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$ $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$	APPROXIMATE: $\beta R_E \geq 10R_2$ $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, V_E = V_B - V_{BE}$ $I_E = \frac{V_E}{R_E}, I_B = \frac{I_E}{\beta + 1}$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$
Collector-feedback		$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C(R_C + R_E)$	
Emitter-follower		$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{EE} - I_E R_E$	
Common-base		$I_E = \frac{V_{EE} - V_{BE}}{R_E}$ $I_B = \frac{I_E}{\beta + 1}, I_C = \beta I_B$ $V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$ $V_{CB} = V_{CC} - I_C R_C$	

BJT Transistor Amplifiers Including effect of R_s and R_L

9	Find I_E , I_B and I_C . (β was not given ☹)	MIST M.Sc.- 2015
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Solution:

Let, $\beta = 90$.

$$V_{EE} - I_E R_E - V_{EB} - I_B R_B - V_{BB} = 0$$

$$\Rightarrow V_{EE} - (\beta + 1) I_B R_E - V_{EB} - I_B R_B - V_{BB} = 0 \quad [\because I_E = (\beta + 1) I_B]$$

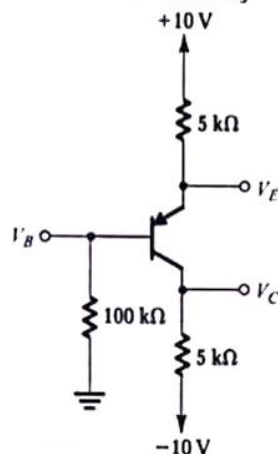
$$\Rightarrow (\beta + 1) I_B R_E + I_B R_B = V_{EE} - V_{EB} - V_{BB}$$

$$\Rightarrow I_B = \frac{V_{EE} - V_{EB} - V_{BB}}{(\beta + 1) R_E + R_B} = \frac{10 - 0.7 - 2.5}{(90 + 1) \times (6.8 \times 10^3) + (100 \times 10^3)} = 9.46 \mu A$$

$$\therefore I_C = \beta I_B = 90 \times 9.46 \mu A = 851.419 \mu A \quad \text{Ans.}$$

$$\therefore I_E = I_B + I_C = 9.46 + 851.419 = 860.879 \mu A \quad \text{Ans.}$$

18	In the circuit shown in Fig., measurement indicates V_B to be +1.0 V and V_E to be +1.7 V. What are α and β for this transistor? What voltage V_C do you expect at the collector? [Exercise 6.14, Sedra-Smith 6 th Edition]	EGCB-20, BUET MSC- 14, BUET M.Sc.- 12
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Solution:

$$V_B = 1 \text{ V}, V_E = 1.7 \text{ V}, \alpha = ?, \beta = ?, V_C = ?$$

$$I_E = \frac{V_{EE} - V_E}{R_E} = \frac{10 - 1.7}{5} = 1.66 \text{ mA}$$

$$I_B = \frac{V_B}{R_B} = \frac{1}{100} = 0.01 \text{ mA}$$

$$I_C = I_E - I_B = 1.66 - 0.01 = 1.65 \text{ mA}$$

$$\beta + 1 = \frac{I_E}{I_B} = \frac{1.66}{0.01} = 166$$

$$\beta = 165$$

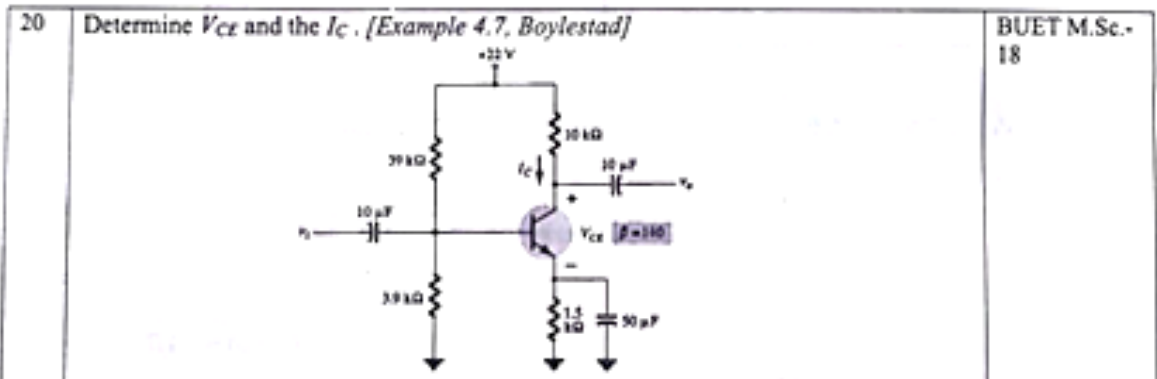
$$\alpha = \frac{\beta}{\beta + 1} = \frac{165}{166} = 0.994$$

$$V_C = V_{CC} + I_C R_C = -10 + (1.65 \times 5) = -1.75 \text{ V}$$

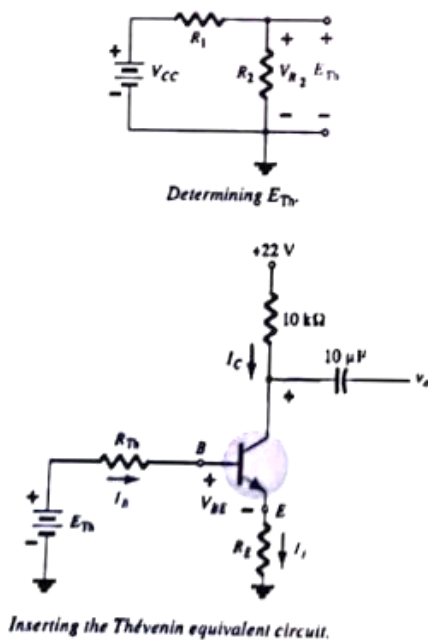
Ans.

Ans.

Ans.



Solution:



$$R_{Th} = R_1 \parallel R_2 = \frac{(39 \text{ k}\Omega)(3.9 \text{ k}\Omega)}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 3.55 \text{ k}\Omega$$

$$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{(3.9 \text{ k}\Omega)(22 \text{ V})}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 2 \text{ V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2 \text{ V} - 0.7 \text{ V}}{3.55 \text{ k}\Omega + (141)(1.5 \text{ k}\Omega)} = \frac{1.3 \text{ V}}{3.55 \text{ k}\Omega + 211.5 \text{ k}\Omega} = 6.05 \mu\text{A}$$

$$I_C = \beta I_B = (140)(6.05 \mu\text{A}) = 0.85 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 22 \text{ V} - (0.85 \text{ mA})(10 \text{ k}\Omega + 1.5 \text{ k}\Omega) = 22 \text{ V} - 9.78 \text{ V} = 12.22 \text{ V}$$

37	Determine R_B and R_C for the transistor inverter, if $I_{C_{sat}} = 10 \text{ mA}$. [Example 4.24, Boylestad]	BUET M.Sc.-15, [Example 4.24, Boylestad]
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Solution:

At saturation: $I_{C_{sat}} = \frac{V_{CC}}{R_C}$
 $10 \text{ mA} = \frac{10 \text{ V}}{R_C}$
 $R_C = \frac{10 \text{ V}}{10 \text{ mA}} = 1 \text{ k}\Omega$

At saturation: $I_B = \frac{I_{C_{sat}}}{\beta_{dc}} = \frac{10 \text{ mA}}{250} = 40 \mu\text{A}$

Choosing $I_B = 60 \mu\text{A}$ to ensure saturation and using $I_B = \frac{V_i - 0.7 \text{ V}}{R_B}$

we obtain $R_B = \frac{V_i - 0.7 \text{ V}}{I_B} = \frac{10 \text{ V} - 0.7 \text{ V}}{60 \mu\text{A}} = 155 \text{ k}\Omega$

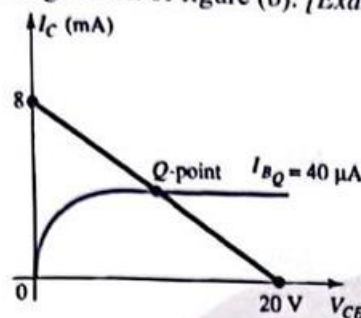
If we choose $R_B = 150 \text{ k}\Omega$, which is a standard value. Then

$$I_B = \frac{V_i - 0.7 \text{ V}}{R_B} = \frac{10 \text{ V} - 0.7 \text{ V}}{150 \text{ k}\Omega} = 62 \mu\text{A}$$

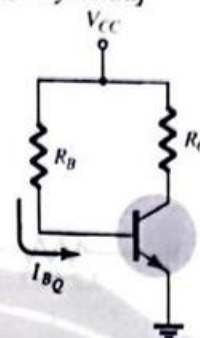
and $I_B = 62 \mu\text{A} > \frac{I_{C_{sat}}}{\beta_{dc}} = 40 \mu\text{A}$

Therefore, we use $R_B = 150 \text{ k}\Omega$ and $R_C = 1 \text{ k}\Omega$.

36	Given the device characteristics of Figure (a). Determine V_{CC} , R_B , and R_C for the fixed-bias configuration of figure (b). [Example 4.19, Boylestad]	NWPGCL-17
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(a)



(b)

Solution:

From the load line

$$V_{CC} = 20 \text{ V}$$

$$I_C = \frac{V_{CC}}{R_C} \bigg|_{V_{CE} = 0 \text{ V}}$$

and $R_C = \frac{V_{CC}}{I_C}$
 $= \frac{20 \text{ V}}{8 \text{ mA}} = 2.5 \text{ k}\Omega$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

with $R_B = \frac{V_{CC} - V_{BE}}{I_B}$
 $= \frac{20 \text{ V} - 0.7 \text{ V}}{40 \mu\text{A}} = \frac{19.3 \text{ V}}{40 \mu\text{A}}$
 $= 482.5 \text{ k}\Omega$

Standard resistor values: $R_C = 2.4 \text{ k}\Omega$

$$R_B = 470 \text{ k}\Omega$$

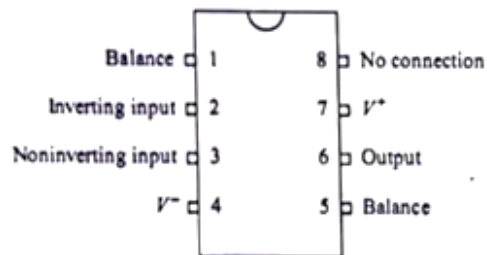
Using standard resistor values gives

$$I_B = 41.1 \mu\text{A}$$

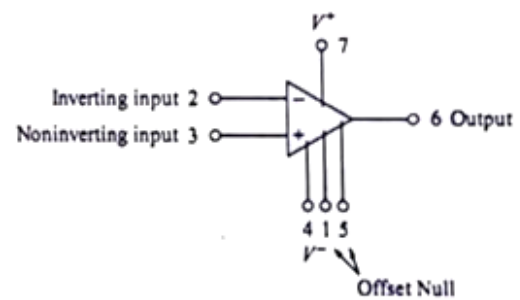
which is well within 5% of the value specified.

The op amp is an electronic unit that behaves like a voltage-controlled voltage source.

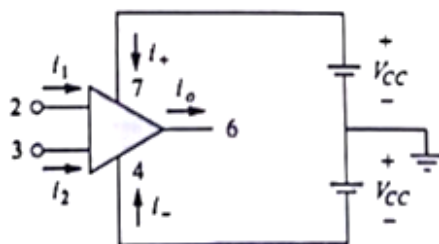
An op amp is an active circuit element designed to perform mathematical operations of addition, subtraction, multiplication, division, differentiation, and integration.



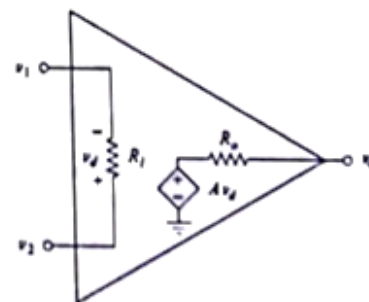
A typical op amp: (a) pin configuration



(b) circuit symbol.



Powering the op amp.

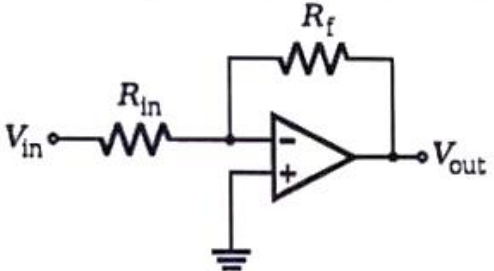
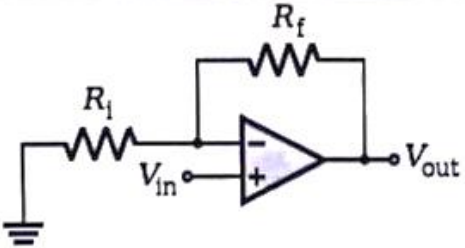
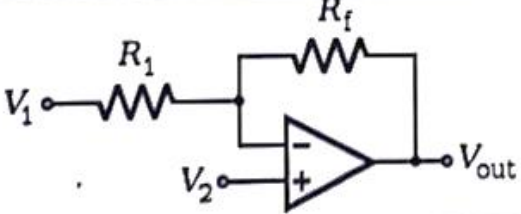


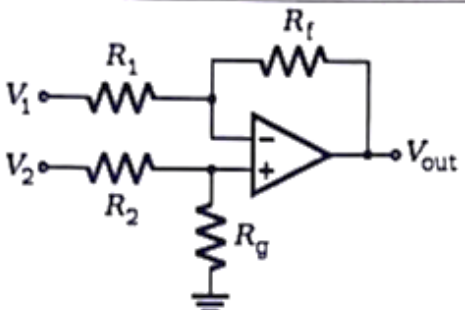
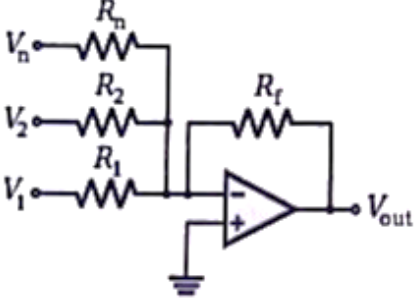
The equivalent circuit of the nonideal op amp.

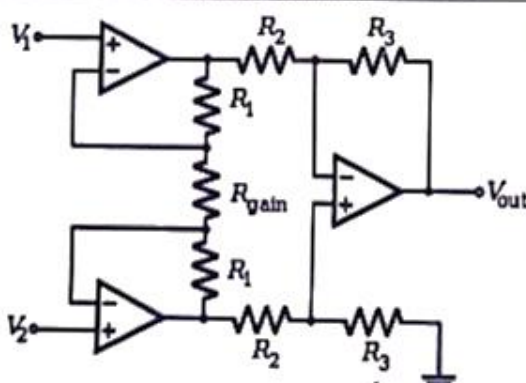
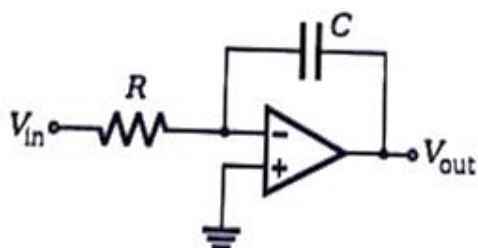
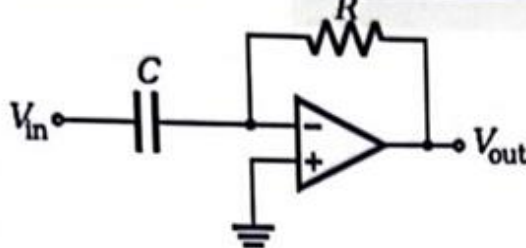
Characteristics of ideal Op-Amp:

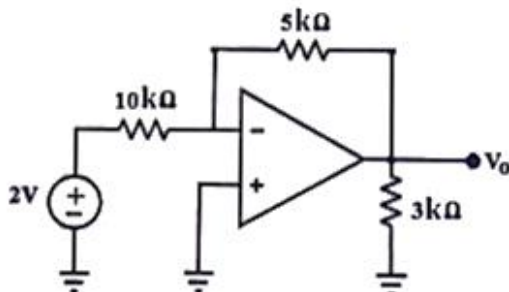
1. Infinite Open-Loop Gain
2. Infinite Input Impedance
3. Infinite Bandwidth
4. Zero Output Impedance
5. Zero Noise Contribution
6. Zero output Offset
7. Differential Inputs Stick Together

Configurations	Characteristics
Voltage follower or buffer	$V_{out} = V_{in}$
Comparator	Compares two voltages and switches its output to indicate which voltage is larger. $V_{out} = \begin{cases} V_{S+} & \text{when } V_1 > V_2 \\ V_{S-} & \text{when } V_1 < V_2 \end{cases}$ (Where, V_S is the supply voltage and the op-amp is powered by $+V_S$ and $-V_S$)

 Inverting amplifier	$V_{out} = -\frac{R_f}{R_{in}} V_{in}$
 Non-inverting amplifier	$V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{in}$
	$V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_2 - \frac{R_f}{R_1} V_1$

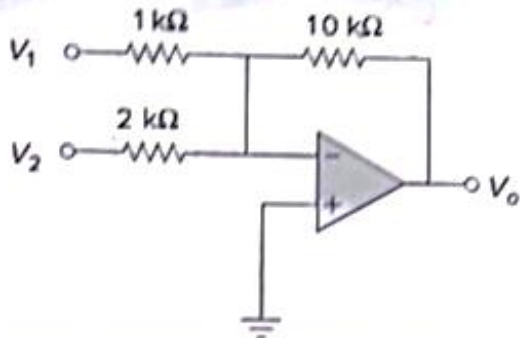
 Differential amplifier	$V_{out} = \frac{R_g}{(R_g + R_2)} \left(1 + \frac{R_f}{R_1}\right) V_2 - \frac{R_f}{R_1} V_1$
 Summing amplifier	$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_n}{R_n} \right)$

 Instrumentation amplifier	$V_{out} = (V_2 - V_1) \left(1 + \frac{2R_1}{R_{gain}} \right) \frac{R_3}{R_2}$
 Inverting Integrator	$V_{out} = -\frac{1}{RC} \int_0^t V_{in} dt + V_{initial}$ (where V_{in} and V_{out} are functions of time, $V_{initial}$ is the output voltage of the integrator at time $t = 0$.) At any single frequency: $v_o = -\frac{v_i}{\omega RC} = -\frac{v_i}{2\pi f RC}$
 Inverting differentiator	$V_{out} = -RC \frac{dV_{in}}{dt}$ At any single frequency: $v_o = -\omega RC v_i = -2\pi f RC v_i$

2	Find the gain in the following op-amp circuit. 	BUET M.Sc.-11
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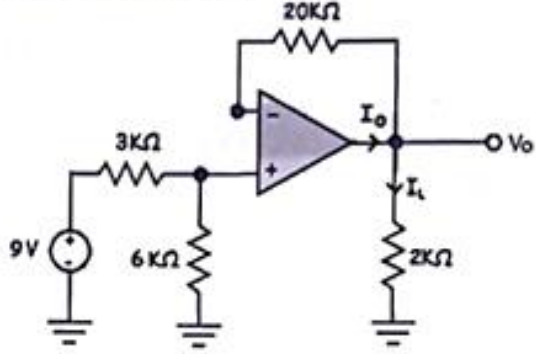
Solution:

$$gain = \frac{V_o}{V_i} = -\frac{R_f}{R_i} = -\frac{5}{10} = -\frac{1}{2}$$

5	Find V_o 	BUET M.Sc.-17
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Solution:

$$V_o = -\frac{R_f}{R_1} \times V_1 - \frac{R_f}{R_2} \times V_2 = -\frac{10}{1} \times V_1 - \frac{10}{2} \times V_2 = -10V_1 - 5V_2 \quad \text{Ans.}$$

41	Find V_o and I_o in the following circuit. 	BUET M.Sc.-16
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Solution:

Applying voltage divider rule, voltage across 6 kΩ resistor,

$$V_x = \left\{ \frac{6}{3+6} \right\} \times 9 = 6 \text{ V}$$

As differential inputs stick together in an op-amp, voltage of inverting (-) input will be equal to voltage of non-inverting (+) input.

As op-amp has infinite input impedance, no current can flow from noninverting (+) input to inverting (-) input. As a result, no current will be flown from noninverting input to output (through 20 kΩ resistor).

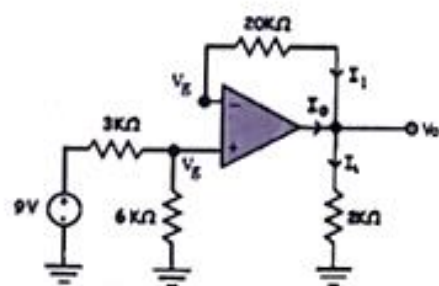
So, Voltage drop across 20 kΩ resistor will be zero. As a result, the output voltage will be equal to voltage of noninverting (+) input.

$$\therefore V_o = V_x = 6 \text{ V} \quad \text{Ans.}$$

$$\therefore i_1 = 0 \text{ A}$$

$$\therefore i_2 = V_o / R_L = 6 / 2 = 3 \text{ mA}$$

$$\therefore i_o = i_2 - i_1 = 3 - 0 = 3 \text{ mA} \quad \text{Ans.}$$

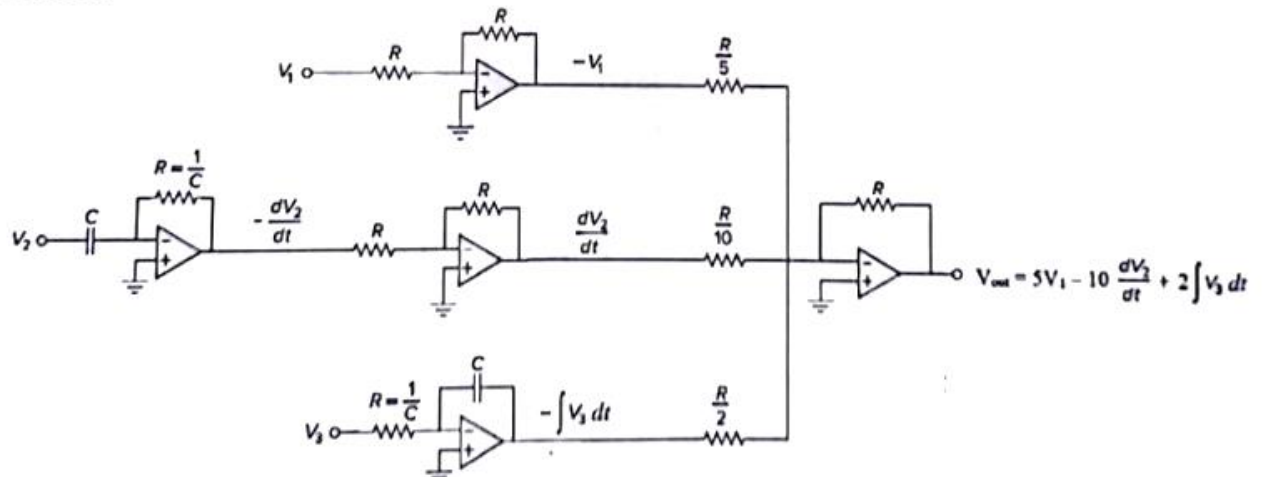


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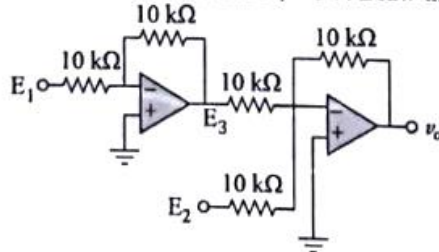
Design using Op-Amp: $V_{out} = 5V_1 - 10 \frac{dV_2}{dt} + 2 \int V_3 dt$

BUET M.Sc.-12

Solution:



50

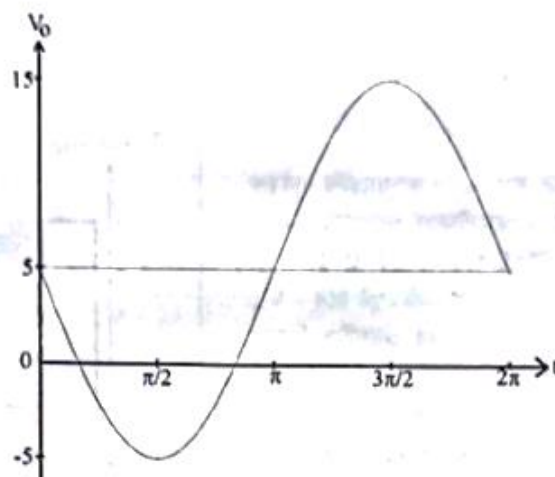
 E_2 is a 10V peak-to-peak sinusoidal source and $E_1 = 5V$. Draw the waveshape of V_O .BUET
M.Sc.-19

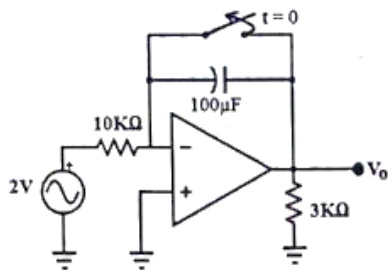
Solution:

$$E_2 = 10 \sin \omega t$$

$$E_3 = -\frac{10}{10} \times E_1 = -E_1 = -5V$$

$$V_O = -\frac{10}{10} \times E_3 - \frac{10}{10} \times E_2 = -\frac{10}{10} \times (-5) - \frac{10}{10} \times E_2 = 5 - E_2 = 5 - 10 \sin \omega t$$



33	Draw the wave shape of the following figure at $t > 0$ and $t < 0$ sec. 	BUET M.Sc.-11
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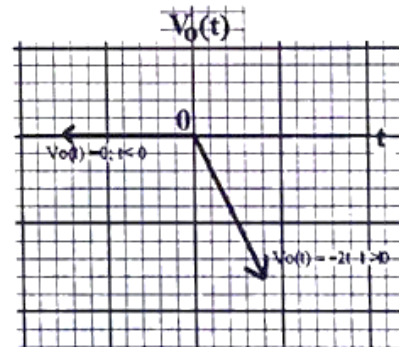
Solution:

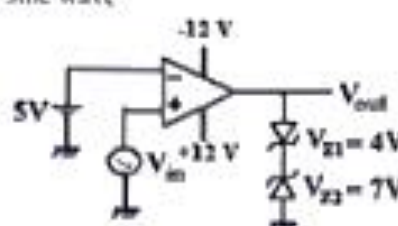
At $t < 0$, switch was closed (connected). So, $V_o = 0$ V then.

At $t > 0$, switch was open (not connected).

So, the circuit becomes an integrator and

$$\begin{aligned}
 v_o &= -\frac{1}{RC} \int v_i dt \\
 &= -\frac{1}{10 \times 10^3 \times 100 \times 10^{-6}} \int 2 dt = -2t
 \end{aligned}$$



52	Draw V_{in} vs V_{out} transfer curve of the following circuit. Biasing voltage of the op-amp is 12V & V_{in} is 10 V peak sine wave 	BUET M.Sc.-18, BUET M.Sc.-14, BUET M.Sc.-17
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ব্যাখ্যাসহ উত্তর: এই প্রশ্নে প্রথমে একটি অপ-অ্যাম্প কন্ফিগারেশনের ব্যবহার করা হয়েছে যা অপ-অ্যাম্প এর দুই পিনের ডিফারেন্সকে কম্পার করবে। পজিটিভ পিনের ডিফারেন্স বেশি হলে আউটপুট হবে $+V_{sat}$ (এক্ষেত্রে +12 ভোল্ট) এবং নেগেটিভ পিনের ডিফারেন্স বেশি হলে আউটপুট হবে $-V_{sat}$ (এক্ষেত্রে -12 ভোল্ট)। এই অর্থে অপ-অ্যাম্প এর নেগেটিভ পিনের ডিফারেন্স ০ ভোল্ট ডিফারেন্স, আর পজিটিভ পিনের ডিফারেন্স ১০ ভোল্ট পিনের একটি সাইন ওয়েভ (চিত্র-১)। তাই যখনই সাইন ওয়েভের মান ০ ভোল্টের বেশি হবে তখনই অপ-অ্যাম্প এর আউটপুট $+V_{sat}$ (অর্থাৎ +12 ভোল্ট) হবে। আর যখন তা ০ ভোল্টের কম হবে তখন আউটপুট $-V_{sat}$ (অর্থাৎ -12 ভোল্ট) হবে (চিত্র-২)।

এবং, অপ-অ্যাম্প এর আউটপুটকে জেনার ডায়োডসের ইনপুট হিসেবে ব্যবহার করা হয়েছে। এখানে ব্যবহৃত জেনার ডায়োডগুলোকে আইডিয়াল ডায়োড হিসেবে ধরা হয়েছে যার ফরওয়ার্ড ভোল্টেজ ড্রপ শূন্য। জেনার ডায়োডের বৈশিষ্ট্য হচ্ছে যে, এতে রিভার্স ব্যাসে V_Z এর চেয়ে বেশি ভোল্টেজ (এক্ষেত্রে 12 ভোল্ট) দেওয়া হলে তা এর একদিকে ভোল্টেজ V_Z ই রাখে। আর ফরওয়ার্ড ব্যাসে এটি সাধারণ ডায়োডের মতই কাজ করে।

এক্ষেত্রে যখন জেনার ডায়োডসের ইনপুট হিসেবে +12 ভোল্ট দেওয়া হয়েছে, তখন প্রথম জেনার ডায়োড (Z_1) ফরওয়ার্ড এবং দ্বিতীয় জেনার ডায়োড (Z_2) রিভার্স ব্যাসে থাকে। তাই ইনপুট ভোল্টেজ (+12 ভোল্ট) দ্বিতীয় জেনার ডায়োডের জেনার ভোল্টেজ ($V_{Z2} = +9$ ভোল্ট) এর বেশি হবার কারণে আউটপুট V_{Z2} (+9 ভোল্ট) হয়।

আবার যখন জেনার ডায়োডসের ইনপুট হিসেবে -12 ভোল্ট দেওয়া হয়েছে, তখন দ্বিতীয় জেনার ডায়োড (Z_2) ফরওয়ার্ড এবং প্রথম জেনার ডায়োড (Z_1) রিভার্স ব্যাসে থাকে। এক্ষেত্রে ইনপুট ভোল্টেজ (-12 ভোল্ট) নেগেটিভ দিকে প্রথম জেনার ডায়োডের জেনার ভোল্টেজ V_{Z1} (-8 ভোল্ট) এর বেশি হবার কারণে আউটপুট নেগেটিভ দিকে V_{Z1} (অর্থাৎ -8 ভোল্ট) করে দেয় (চিত্র-৩)। এটিই আইডিয়াল আউটপুট। ১ সাইকেল পরে আইডিয়াল আউটপুটের ওয়েভশেপের ফরম হয়, সেটাই ট্রান্সফার ক্যারাক্টারিস্টিক এর ওয়েভশেপ (চিত্র-৪)।

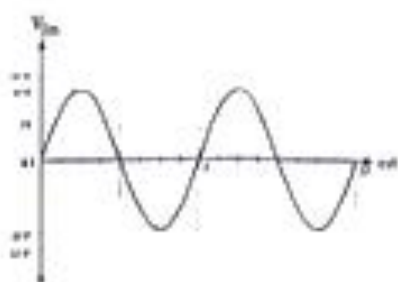


Figure-1: Input sine wave, $V_{in} = 10 \sin \omega t$

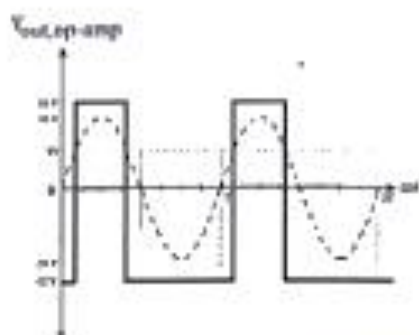


Figure-2: Output of Op-Amp - Input to Zener Diodes

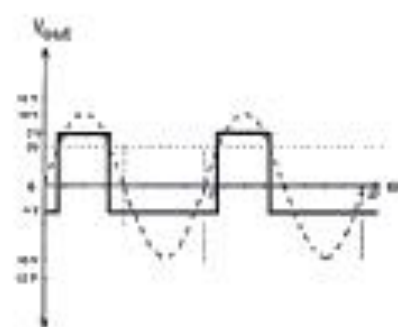


Figure-3: Final Output (after Zener Diodes)

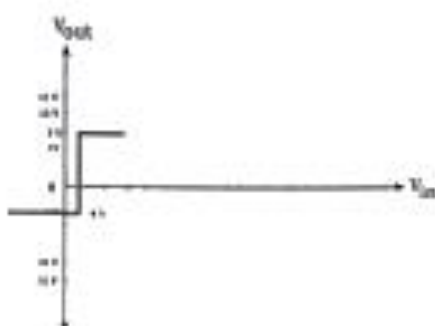


Figure-4: Transfer Characteristics (V_{in} vs V_{out} curve)

$i_D = I_o(e^{v_D/\eta V_T} - 1)$	Where,
When $V_D \gg \eta V_T$,	i_D = Current through in Ampere
$i_D = I_o e^{v_D/\eta V_T}$	V_D = diode terminal voltage in volt
	I_o = temperature-dependent saturation current in Ampere
	V_T = absolute temperature of p-n junction in Kelvin 25 mV in room temperature
	k = Boltzmann's constant = 1.38×10^{-23} J/K
	q = electron charge = 1.6×10^{-19} C
	η = empirical constant = 1 for Ge and 2 for Si

1	For a semiconductor Diode, if forward current is 10mA at $V_d = 0.3V$, then what is the forward current at $V_d = 0.4V$.	DWASA-14
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Solution:

$i_D = I_o(e^{v_D/\eta V_T} - 1)$	Where,
When $V_D \gg \eta V_T$,	V_D = diode terminal voltage in volt
$i_D = I_o e^{v_D/\eta V_T}$	I_o = temperature-dependent saturation current in Ampere
	V_T = absolute temperature of p-n junction in Kelvin
	k = Boltzmann's constant = 1.38×10^{-23} J/K
	q = electron charge = 1.6×10^{-19} C
	η = empirical constant = 1 for Ge and 2 for Si

For germanium diode $\eta = 1$ &

$$i_D = I_o e^{\frac{v_D}{V_T}}$$

$$\therefore \frac{i_{D_2}}{i_{D_1}} = \frac{I_o e^{\frac{v_{D_2}}{V_T}}}{I_o e^{\frac{v_{D_1}}{V_T}}} = e^{\frac{v_{D_2} - v_{D_1}}{V_T}}$$

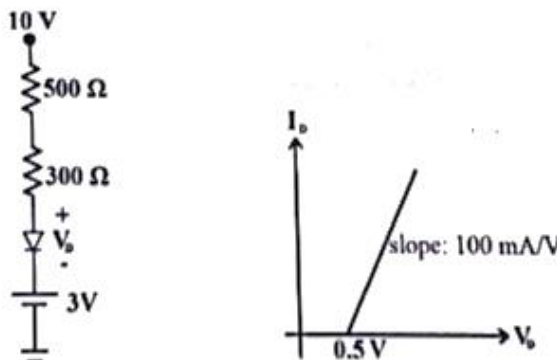
$$\text{or, } i_{D_2} = i_{D_1} e^{\frac{v_{D_2} - v_{D_1}}{V_T}}$$

$$= 10 * e^{\{(0.4-0.3)/0.025\}}$$

$$= 10 * e^4$$

$$= 10 * 54.598 = 545.98 \text{ mA } \underline{\text{Ans.}}$$

Here,	
η	= 1, for germanium
V_{D1}	= 0.3 V at
I_{D1}	= 10 mA
V_{D2}	= 0.4 V
I_{D2}	= ?
V_T	= 25 mV = 0.025 V

8	The diode has following I-V characteristics and it's reverse saturation current is 10 nA. Find I_D. 	BUET M.Sc.-18
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Solution:

Using piecewise linear model,

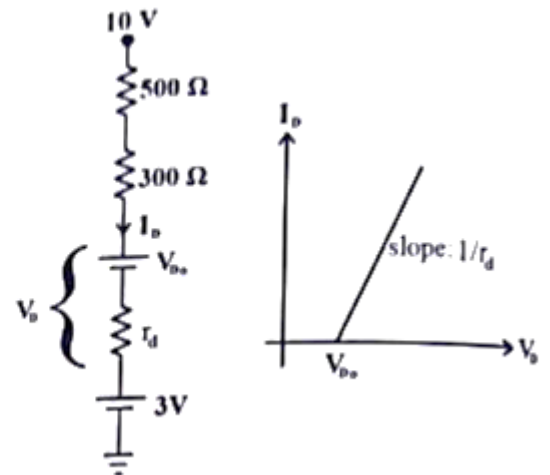
$$V_{D0} = 0.5 \text{ V}$$

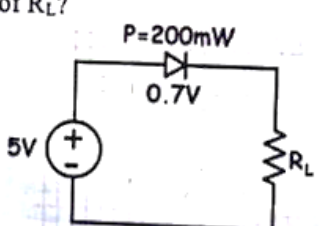
$$\text{Slope, } \frac{1}{r_d} = 100 \text{ mA/V} = 0.1 \text{ A/V}$$

$$\Rightarrow r_d = \frac{1}{0.1} = 10 \text{ V/A} = 10 \Omega$$

$$\therefore I_D = \frac{10 - V_{D0} - 3}{500 + 300 + r_d} = \frac{10 - 0.5 - 3}{500 + 300 + 10} = 8.125 \text{ mA}$$

Ans.



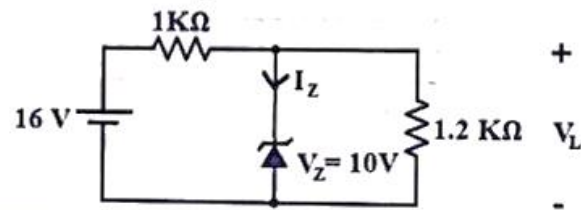
17	What is the maximum value of R_L? 	BUET M.Sc.-15
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Solution:

$$I_D = P/V_D = 200/0.7 = 285.71 \text{ mA}$$

$$\text{So, } R_L = (5 - 0.7) / 285.71 = 0.01505 \text{ k}\Omega = 15.05 \text{ k}\Omega \text{ Ans.}$$

22	Find out V_L and P_Z for the following circuit.	PGCL-11
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Solution:

$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 8.73 \text{ V}$$

Since $V = 8.73 \text{ V}$ is less than $V_Z = 10 \text{ V}$, the diode is in the "off" state

$$V_L = V = 8.73 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 8.73 \text{ V} = 7.27 \text{ V}$$

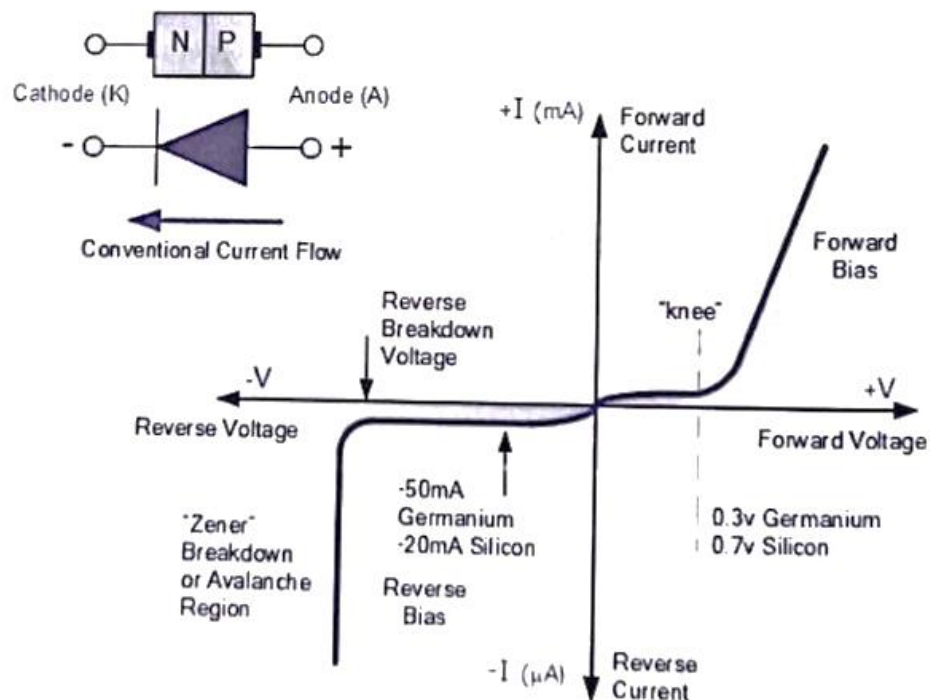
$$I_Z = 0 \text{ A}$$

and

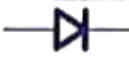
$$P_Z = V_Z I_Z = V_Z (0 \text{ A}) = 0 \text{ W}$$

60	Draw the v-i characteristics of practical Si diode at forward bias and reverse bias.	BPDB-10
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Solution:



Solution:

Parameters	Diode	Zener Diode
Definition	Diode is a semiconductor device which conducts only in forward bias.	Zener diode is semiconductor device which can conduct in forward as well as reversed biased.
Operation in Reverse Bias	It gets damaged in reverse bias.	It can operate without getting damaged.
Circuit Symbol		
Doping Intensity	Doping intensity is low.	Doping intensity is high to achieve sharp breakdown.
Application	Diode is used in rectifiers, clippers, clampers etc.	Zener diode is mostly used in voltage regulator.

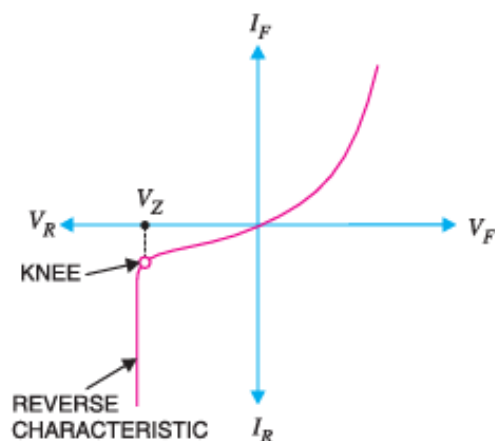


Fig. 6.52

a zener diode.

A properly doped crystal diode which has a sharp breakdown voltage is known as a **zener diode**.

Fig. 6.53 shows the symbol of a zener diode. It may be seen that it is just like an ordinary diode except that the bar is turned into z-shape. The following points may be noted about the zener diode:

(i) A zener diode is like an ordinary diode except that it is properly doped so as to have a sharp breakdown voltage.

(ii) A zener diode is always reverse connected i.e. it is always reverse biased.

(iii) A zener diode has sharp breakdown voltage, called zener voltage V_Z .

(iv) When forward biased, its characteristics are just those of ordinary diode.

(v) The zener diode is not immediately burnt just because it has entered the breakdown region. As long as the external circuit connected to the diode limits the diode current to less than **burn out** value, the diode will not burn out.



Example 6.30. The circuit of Fig. 6.66 uses two zener diodes, each rated at 15 V, 200 mA. If the circuit is connected to a 45-volt unregulated supply, determine :

(i) The regulated output voltage

(ii) The value of series resistance R

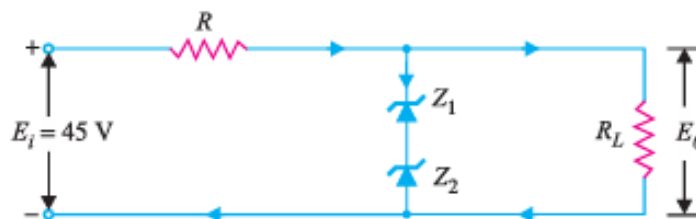


Fig. 6.66

Solution. When the desired regulated output voltage is higher than the rated voltage of the zener, two or more zeners are connected in series as shown in Fig. 6.66. However, in such circuits, care must be taken to select those zeners that have the same current rating.

Current rating of each zener, $I_Z = 200 \text{ mA}$

Voltage rating of each zener, $V_Z = 15 \text{ V}$

Input voltage, $E_i = 45 \text{ V}$

(i) Regulated output voltage, $E_o = 15 + 15 = 30 \text{ V}$

(ii) Series resistance, $R = \frac{E_i - E_o}{I_Z} = \frac{45 - 30}{200 \text{ mA}} = \frac{15 \text{ V}}{200 \text{ mA}} = 75 \Omega$

Example 6.33. In the circuit shown in Fig. 6.69, the voltage across the load is to be maintained at 12 V as load current varies from 0 to 200 mA. Design the regulator. Also find the maximum wattage rating of zener diode.

Solution. By designing the regulator here means to find the values of V_Z and R . Since the load voltage is to be maintained at 12 V, we will use a zener diode of zener voltage 12 V i.e.,

$$V_Z = 12 \text{ V}$$

The voltage across R is to remain constant at $16 - 12 = 4 \text{ V}$ as the load current changes from 0 to 200 mA. The minimum zener current will occur when the load current is maximum.

$$\therefore R = \frac{E_i - E_0}{(I_Z)_{\min} + (I_L)_{\max}} = \frac{16 - 12}{(0 + 200) \text{ mA}} = \frac{4 \text{ V}}{200 \text{ mA}} = 20 \Omega$$

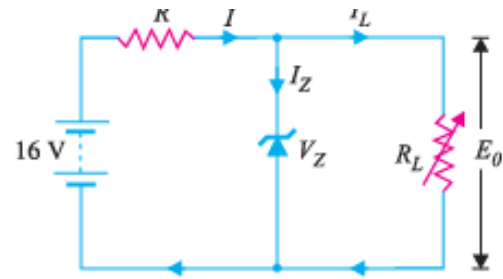


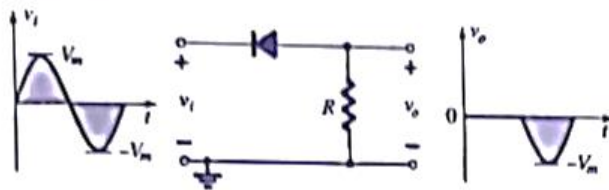
Fig. 6.69

Condition	Circuit with Zener diode	Complete Zener equivalent circuit	Approximate Zener equivalent circuit
$V_i > V_Z$			
$0.7 < V_i < V_Z$			
$0.7 < V_i$			

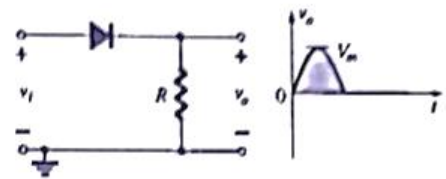
Summary of Clipper Circuits

Simple Series Clippers (Ideal Diodes)

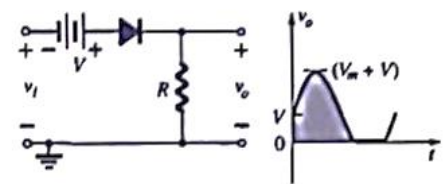
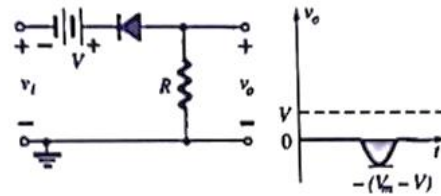
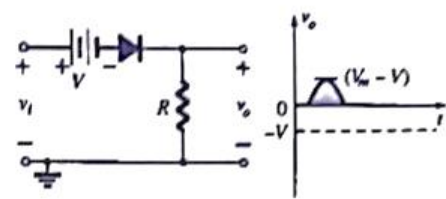
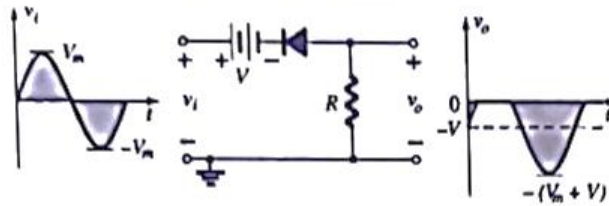
POSITIVE



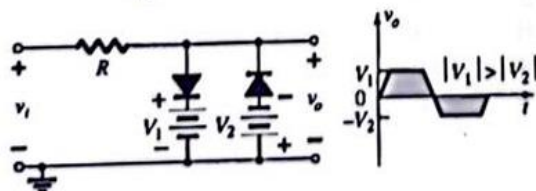
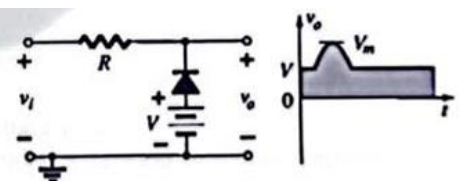
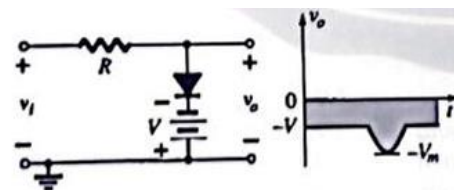
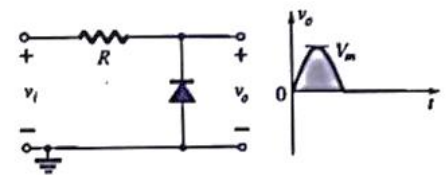
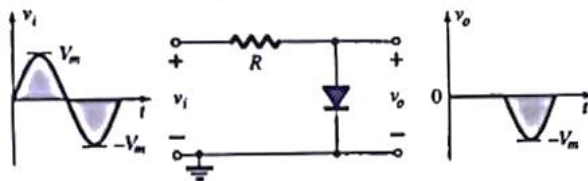
NEGATIVE



Biased Series Clippers (Ideal Diodes)

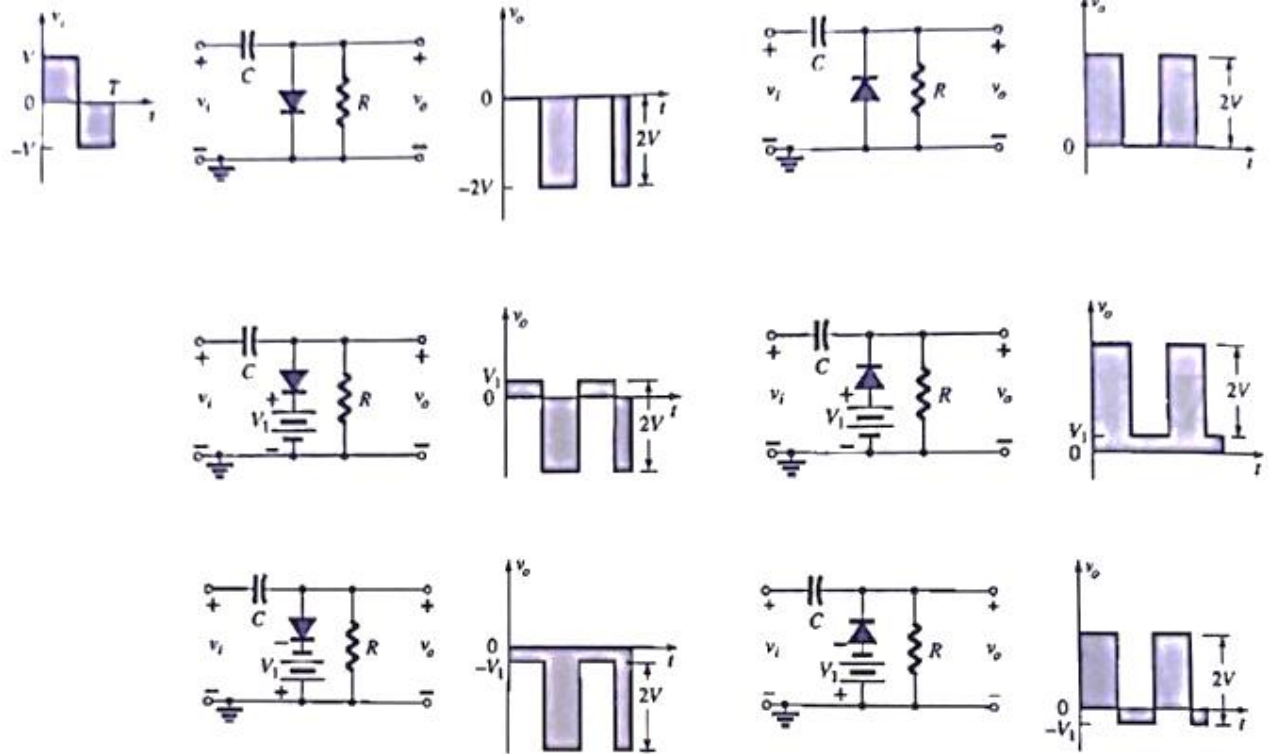


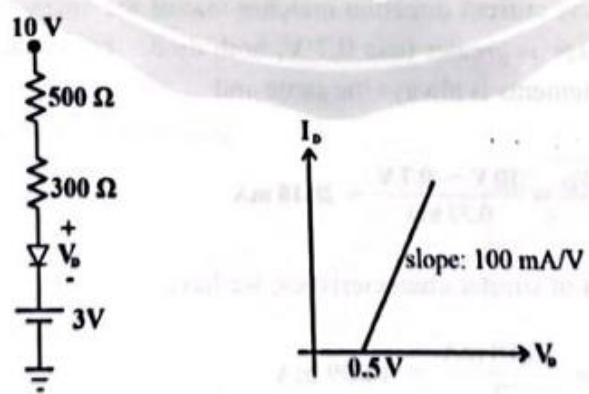
Simple Parallel Clippers (Ideal Diodes)



Summary of Clampers

Clamping Networks



8	The diode has following I-V characteristics and it's reverse saturation current is 10 nA. Find I_D. 	BUET M.Sc.-18
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Solution:

Using piecewise linear model,

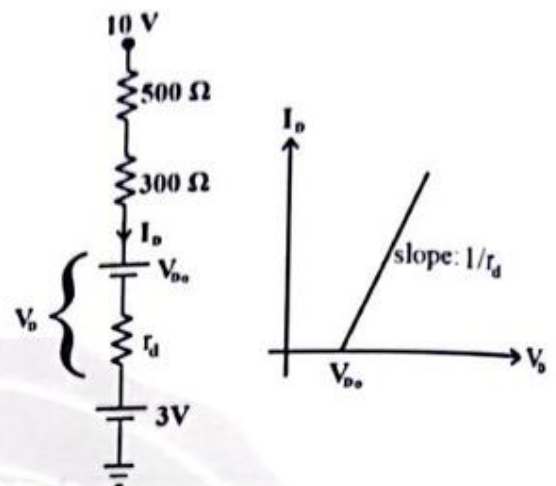
$$V_{D0} = 0.5 \text{ V}$$

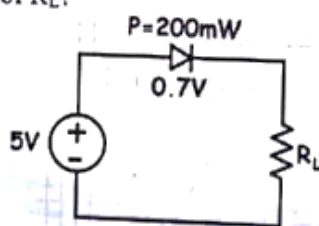
$$\text{Slope, } \frac{1}{r_d} = 100 \text{ mA/V} = 0.1 \text{ A/V}$$

$$\Rightarrow r_d = \frac{1}{0.1} = 10 \text{ V/A} = 10 \Omega$$

$$\therefore I_D = \frac{10 - V_{D0} - 3}{500 + 300 + r_d} = \frac{10 - 0.5 - 3}{500 + 300 + 10} = 8.125 \text{ mA}$$

Ans.



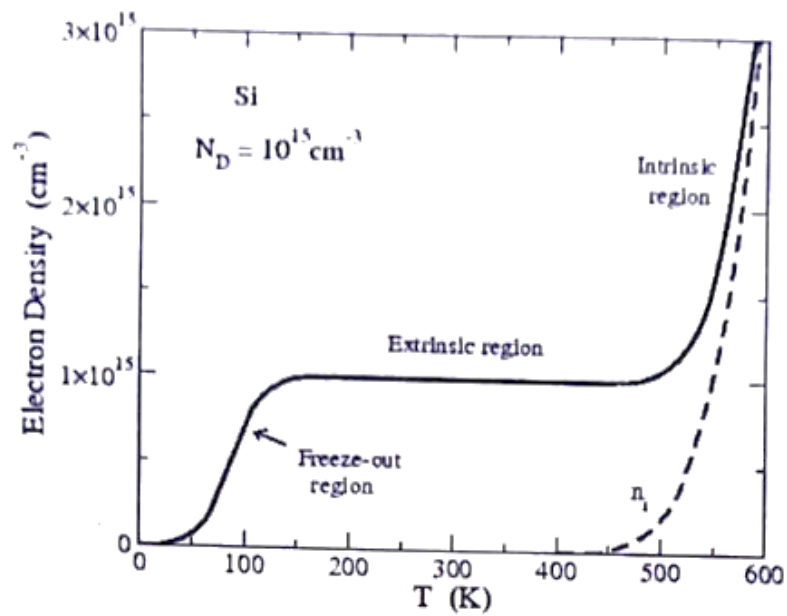
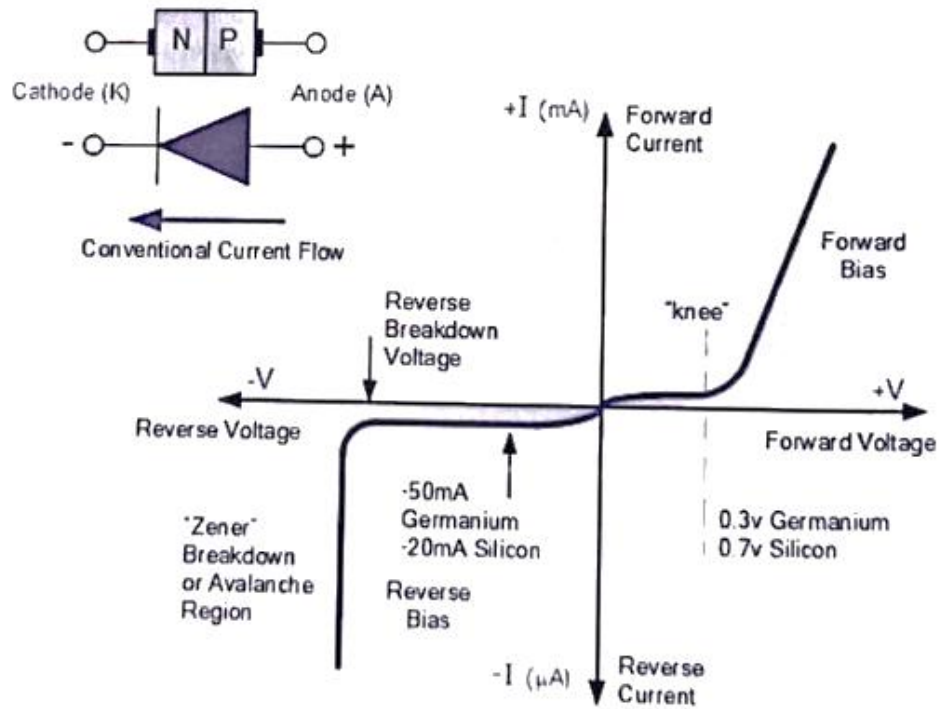
17	What is the maximum value of R_L? 	BUET M.Sc.-15
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Solution:

$$I_D = P/V_D = 200/0.7 = 285.71 \text{ mA}$$

$$\text{So, } R_L = (5 - 0.7) / 285.71 = 0.01505 \text{ k}\Omega = 15.05 \text{ k}\Omega \text{ Ans.}$$

Solution:



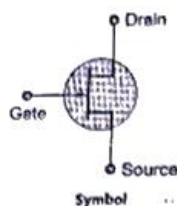
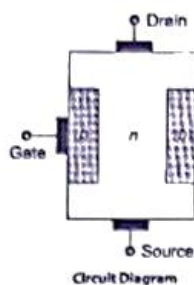
Solution:

Parameters	Diode	Zener Diode
Definition	Diode is a semiconductor device which conducts only in forward bias.	Zener diode is semiconductor device which can conduct in forward as well as reversed biased.
Operation in Reverse Bias	It gets damaged in reverse bias.	It can operate without getting damaged.
Circuit Symbol		
Doping Intensity	Doping intensity is low.	Doping intensity is high to achieve sharp breakdown.
Application	Diode is used in rectifiers, clippers, clampers etc.	Zener diode is mostly used in voltage regulator.

FET

- FET is a unipolar and voltage controlled device
- The terminals drain, Gate and source of a FET are identical to collector base and emitter of a BJT.
- Since the input function is reverse biased in JFET, the current drawn is very small and so input impedance is very high.
- Less noisy device due to absence of minority carriers.
- Excellent thermal stability due to absence of leakage current.
- FET is considered as excellent signal chopper because of zero offset voltage

Circuit Diagram, Symbol and Formula



$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2 \text{ in saturation region}$$

Where,

I_D = Drain Current

I_{DSS} = Maximum value of current when $V_{GS} = 0$

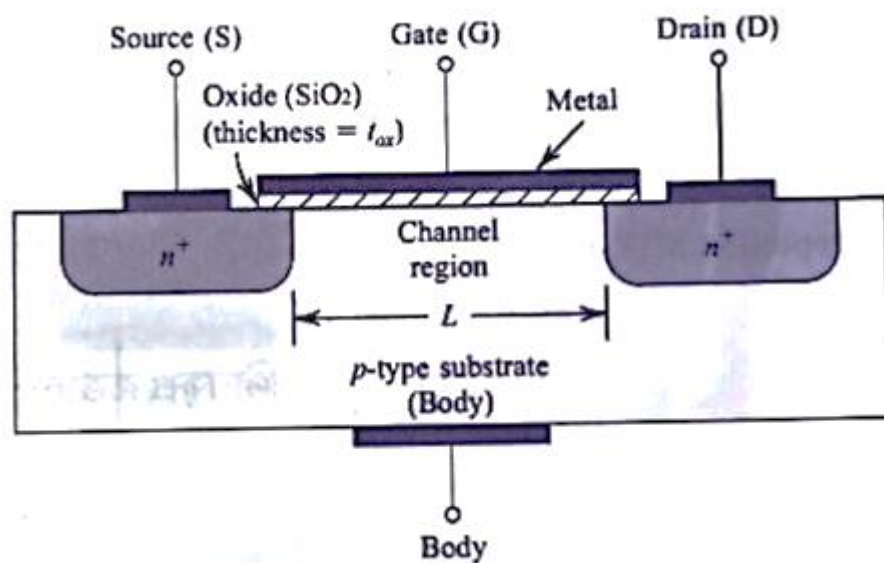
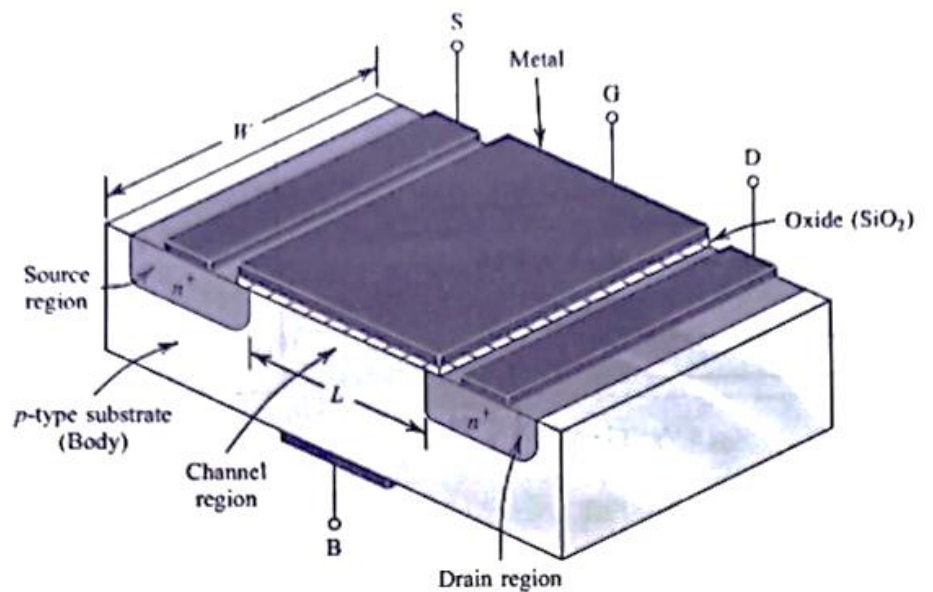
V_p = Pinch off voltage

V_{GS} = Gate to source voltage

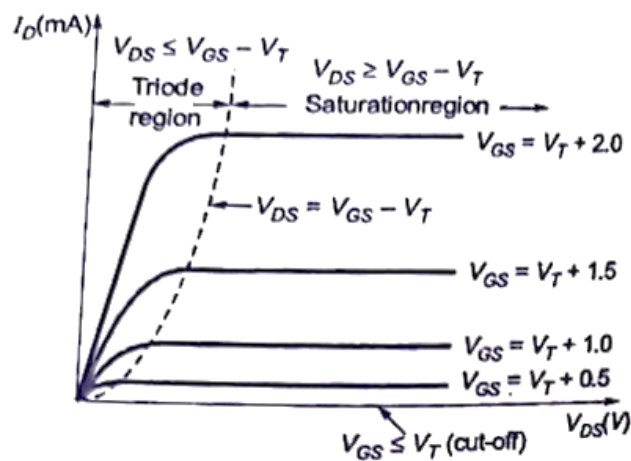
MOSFET

- MOSFET is an integrated device, fabricated by VLSI using planner technology.
- It is a voltage controlled device.
- In the MOSFET the plate and semiconductor channel will be working as the plate of capacitor and SiO_2 as dielectric.
- The large input impedance of MOSFET is due to the SiO_2 .
- MOSFET is very sensitive to static electrical noise and static electrical disturbances.

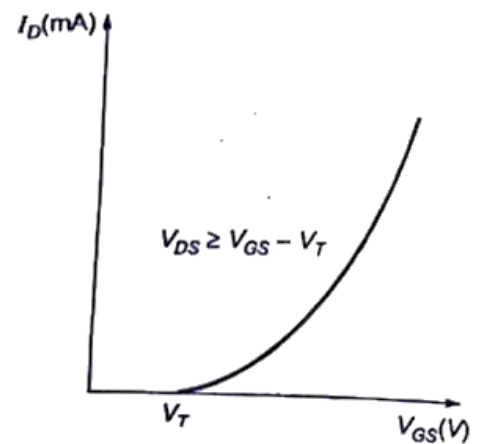
Structure of MOSFET:



Characteristics of MOSFET



Drain Characteristic of E-MOSFET



Transfer Characteristic of E-MOSFET

- Threshold voltage (V_t) is defined as the "Minimum gate to source voltage required for MOSFET turn into ON state".
- For better performance of the MOSFET, threshold voltage must be as small as possible.
- The threshold voltage of MOSFET can be graphically obtained from the transfer characteristics.
- Threshold voltage V_T should be smaller.

For depletion MOSFET

$$\text{Drain current, } I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_p} \right]^2$$

Where, $I_{DSS} = I_{Dmax}$

- In Depletion MOSFET channel is pre-existing.

For enhancement type n-channel MOSFET

$V_{GS} > V_t$	for 'ON' or saturation region
$V_{GS} < V_t$	for 'OFF' or cut-off region
$V_{DS} < (V_{GS} - V_t)$	for Triode region of operation
$V_{DS} \geq (V_{GS} - V_t)$	for Saturation region of operation

- For saturation region, $I_D = K (V_{GS} - V_t)^2$
- For P-channel MOSFET $I_D = K (V_{GS} - V_t)^2$
- In Enhancement MOSFET Channel is diffused channel.

অংকের জন্য প্রয়োজনীয় সূত্রসমূহ (For enhancement type n-channel MOSFET)

Condition	Operation Region	Formula of I_D
$V_{DS} < (V_{GS} - V_t)$	Triode region	$I_D = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2$
$V_{DS} \geq (V_{GS} - V_t)$	Saturation region	$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$

Where,

μ_n = mobility of the electrons at the surface of the channel

C_{ox} = oxide capacitance = $\frac{\epsilon_{ox}}{t_{ox}}$

W = Channel width

L = Channel Length

V_{GS} = Gate to Source Voltage

V_{DS} = Drain to Source Voltage

V_t = Threshold voltage

= Minimum gate to source voltage required for turning MOSFET into ON state

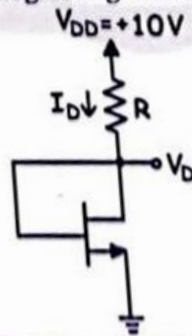
K_n' = process trans-conductance parameter = $\mu_n C_{ox}$

K_n = MOSFET trans-conductance parameter = $\mu_n C_{ox} \frac{W}{L}$

$\frac{W}{L}$ = aspect ratio

2

In the following NMOS, $I_D = 0.4 \text{ mA}$, $V_t = 2 \text{ V}$, $\mu_n C_{ox} = 20 \mu\text{A/V}^2$, $L = 10 \mu\text{m}$ and $W = 100 \mu\text{m}$. Find the value of R neglecting channel length and modulation effect.



BUET
M.Sc.-18,
BUET
M.Sc.-12

Solution:

Because $V_{DG} = 0$, the FET is operating in the saturation region. Thus

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

$$0.4 = \frac{1}{2} (20)(10^{-6})(100/10)(V_{GS} - 2)^2$$

which yields two values for V_{GS} , namely 4 and 0. The second value obviously does not make physical sense since it is lower than V_t . Thus $V_{GS} = 4 \text{ V}$, and the drain voltage will be

$$V_D = +4 \text{ V}$$

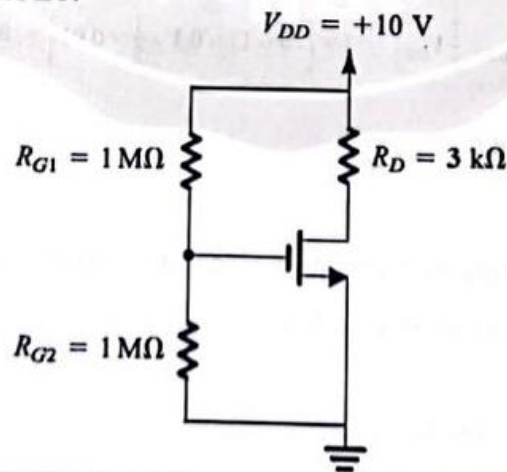
The required value for R can be found as follows:

$$\begin{aligned} R &= \frac{V_{DD} - V_D}{I_D} \\ &= \frac{10 - 4}{0.4} = 15 \text{ k}\Omega \end{aligned}$$

7

Find I_{Dmax} that can flow through the ideal N-MOSFET. Also find V_{DS} at I_{Dmax} . Given $V_{tn} = 1.5$ V and $K_n'(W/L) = 2.76 \times 10^{-6}$ A/V². Also Find the power dissipated in the MOSFET.

RPGCL-17,
BUET
M.Sc.-16



Solution:

Since the gate current is zero, the voltage at the gate is simply determined by the voltage divider formed by the two 1-MΩ resistors

$$V_G = V_{DD} \frac{R_{G2}}{R_{G2} + R_{G1}} = 10 \times \frac{1}{1+1} = +5 \text{ V}$$

With this positive voltage at the gate, the NMOS transistor will be turned on. We do not know, however, whether the transistor will be operating in the saturation region or in the triode region. We shall assume saturation-region operation, solve the problem, and then check the validity of our assumption. If our assumption turns out not to be valid, we will have to solve the problem again for triode-region operation.

Since the voltage at the gate is 5 V and source is grounded,

$$V_{GS} = 5 \text{ V}$$

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 = \frac{1}{2} \times 2.76 \times 10^{-6} \times (5 - 1.5)^2 = 16.9 \text{ } \mu\text{A}$$

$$V_D = V_{DS} = V_{DD} - I_D R_D = 10 - (16.9 \times 10^{-6}) \times (3 \times 10^3) = 9.9493 \text{ V.}$$

Since, $V_{DS} > V_{GS} - V_{tn}$; the transistor is operating in saturation, as initially assumed.

So, $I_{Dmax} = I_D = 16.9 \text{ } \mu\text{A}$ and $V_{DS} = 9.9493 \text{ V.}$ Ans.

Power Dissipation in MOSFET $P_{MOSFET} = V_{DS} I_D = 9.9493 \times 16.9 = 168.14 \text{ mW}$ Ans.

Basic k-Map

A \ B	0	1
0	0	1
1	2	3

(a) Two Variables

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

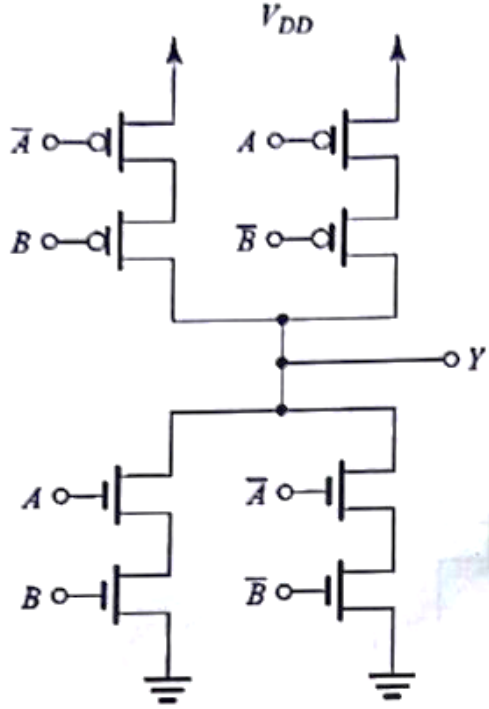
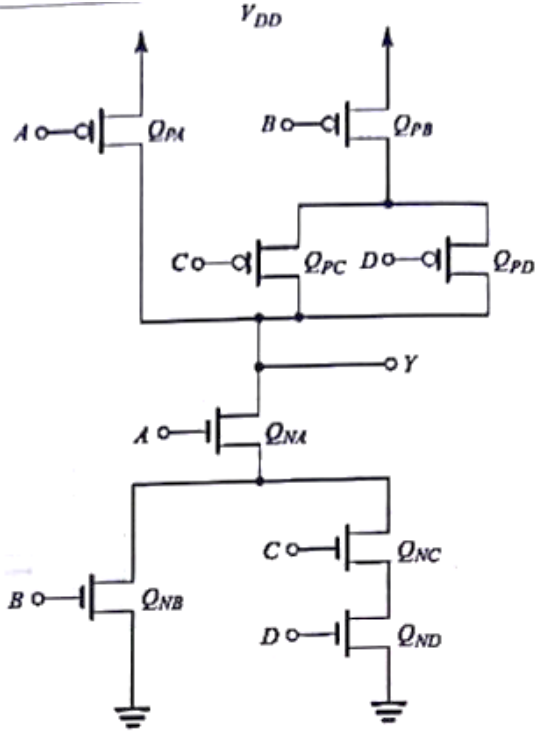
(b) Three Variables

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

(c) Four Variables

Important CMOS Gates

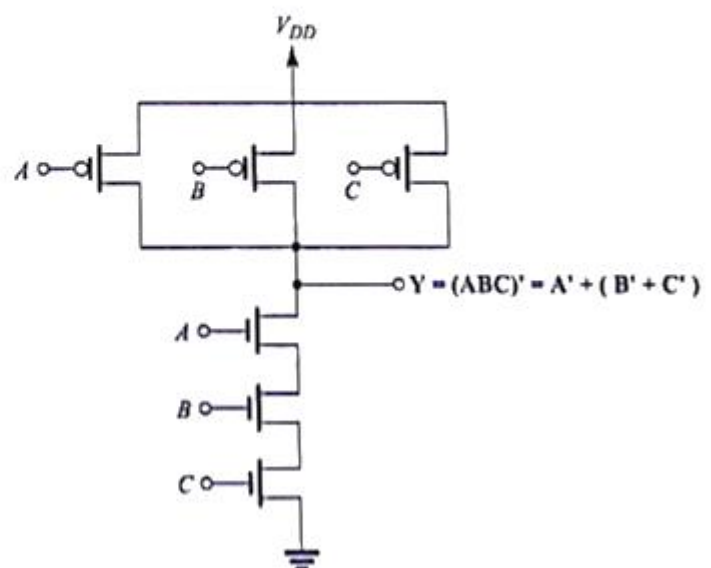
$Y = \overline{A}$	$Y = \overline{A + B}$	$Y = \overline{AB}$
CMOS Inverter	Two-input CMOS NOR gate	Two-input CMOS NAND gate

 $Y = (AB + A'B')' = A \oplus B$	 $Y = \overline{A(B + CD)}$
The complete XOR realization using CMOS Gates	Example: $Y = \{A(B+CD)\}'$ realization using CMOS Gates

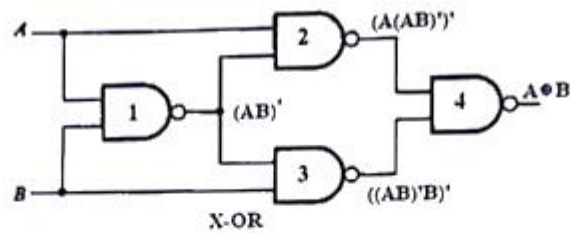
12	Draw the CMOS complex gate logic circuit function on $Y = A' + (B' + C')$	BUET M.Sc.-18, BUET MSC-14
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Solution:

$$\begin{aligned}
 Y &= A' + (B' + C') \\
 \therefore Y' &= \{A' + (B' + C')\}' \\
 &= (A')' \cdot (B' + C')' \\
 &= (A')' \cdot \{(B')' \cdot (C')'\} \\
 &= A B C \\
 \therefore Y &= (Y')' = (A B C)'
 \end{aligned}$$



Solution:



Explanation:

Gate No.	Inputs	Output
1	A, B	$(AB)'$
2	A, $(AB)'$	$(A(AB)')'$
3	$(AB)'$, B	$(B(AB)')'$
4	$(A(AB)')'$, $(B(AB)')'$	$A'B + AB'$

Now the output from gate no. 4 is the overall output of the configuration.

$$\begin{aligned}
 Y &= ((A(AB)')'(B(AB)')')' \\
 &= (A(AB)')'' + (B(AB)')'' \\
 &= (A(AB)') + (B(AB)') \\
 &= (A(A' + B')) + (B(A' + B')) \\
 &= (AA' + AB') + (BA' + BB') \\
 &= (0 + AB' + BA' + 0) \\
 &= AB' + BA' \\
 \Rightarrow Y &= AB' + A'B
 \end{aligned}$$